1. PROCEDURE

1. Create a Python Flask web application containing a simple form with Username and Message fields.
2. Run the Flask application locally on 127.0.0.1:5000.
3. Open Burp Suite Community Edition and start a temporary project.
4. Go to Proxy → Intercept and enable Intercept.
5. Configure the browser to use Burp Suite as the proxy on 127.0.0.1:8080.
6. Open the Flask application in the browser using http://127.0.0.1:5000.
7. Enter a username and message in the form and click Send Request.
8. Observe the HTTP POST request intercepted by Burp Suite.
9. Inspect the request headers and form parameters such as username and message.
10. Modify the username parameter in the intercepted request and click Forward.
11. Observe the modified request being sent to the Flask server.
12. Verify the server response in the browser and check the modified username and message.
13. Use Proxy → HTTP history to view the captured web traffic.

2. SOURCE CODE

File: app.py

Here’s a clean copy-paste version for your documentation:

Flask Application – Burp Suite Traffic Demo

```
from flask import Flask, request


app = Flask(__name__)


HTML = """
<!DOCTYPE html>
<html>
<head>
    <title>SSD Lab - Burp Suite</title>
    <style>
        body {
            font-family: Arial;
            margin: 60px;
            background: #f4f4f4;
        }

        .box {
            width: 400px;
            padding: 25px;
            background: white;
            margin: auto;
            border-radius: 10px;
        }

        input {
            padding: 10px;
            width: 90%;
```

```
            margin: 8px 0;
        }

        button {
            padding: 10px 20px;
        }
    </style>
</head>

<body>
<div class="box">
    <h2>Burp Suite Traffic Demo</h2>

    <form method="POST">
        <input name="username" placeholder="Username">
        <input name="message" placeholder="Message">
        <button type="submit">Send Request</button>
    </form>

    {% if username %}
        <hr>
        <h3>Server Response</h3>
        Username: {{ username }}<br>
        Message: {{ message }}
    {% endif %}
</div>
</body>
</html>
"""

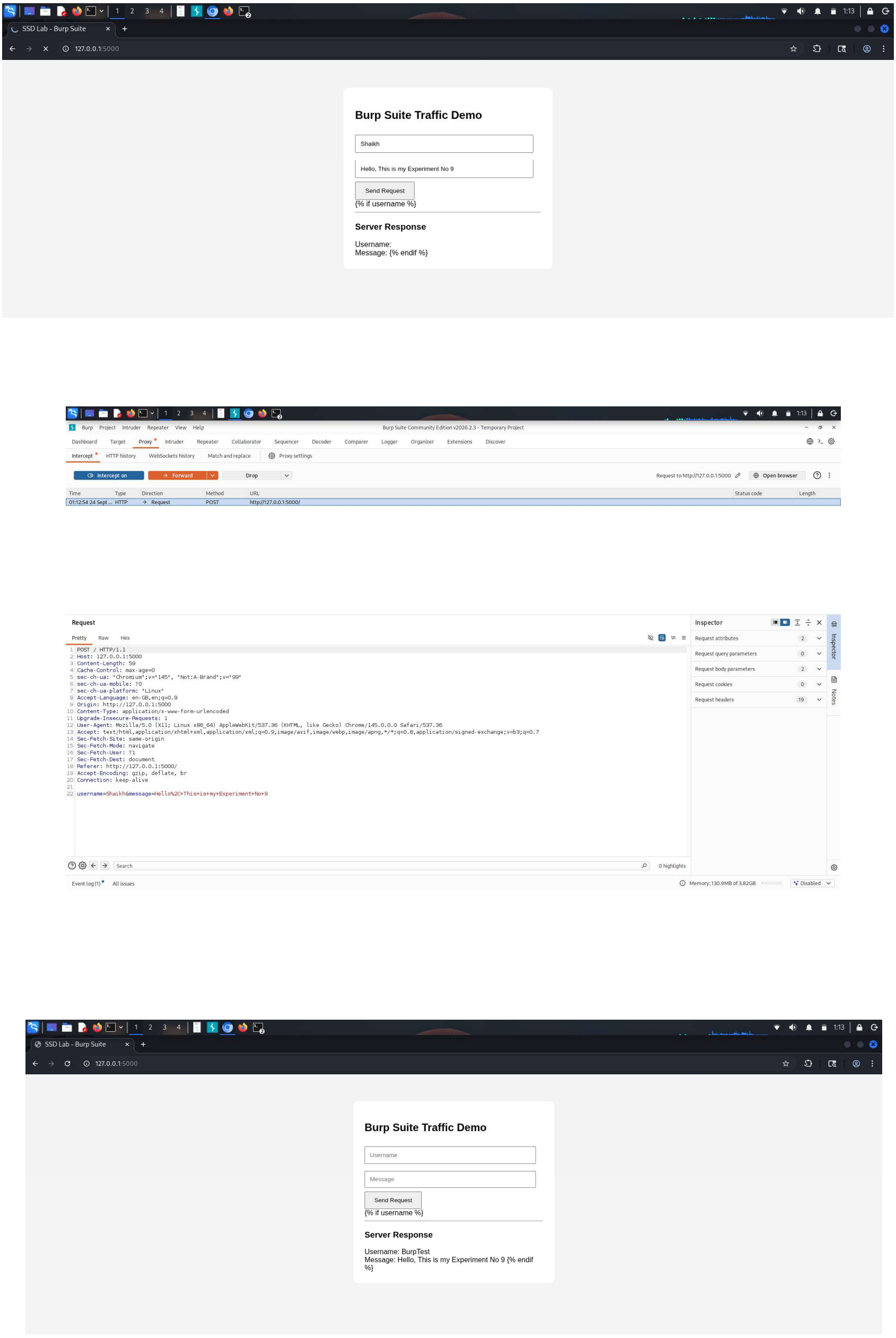
@app.route("/", methods=["GET", "POST"])
def home():
    username = ""
    message = ""

    if request.method == "POST":
        username = request.form.get("username", "")
        message = request.form.get("message", "")

    return HTML.replace("{{ username }}", username).replace(
        "{{ message }}", message
    )


if __name__ == "__main__":
    app.run(host="127.0.0.1", port=5000)
```

3. OUTPUT


SSD Lab - Burp Suite
127.0.0.1:5000
Burp Suite Traffic Demo
Shaikh
Hello, This is my Experiment No 9
Send Request
{% if username %}
Server Response
Username:
Message: {% endif %}
Burp Project Intruder Repeater View Help
Burp Suite Community Edition v2026.2.3 - Temporary Project
Dashboard Target Proxy Intruder Repeater Collaborator Sequencer Decoder Comparer Logger Organizer Extensions Discover
Intercept HTTP history WebSockets history Match and replace Proxy settings
Intercept on
Forward
Drop
Request to http://127.0.0.1:5000
Open browser
Time Type Direction Method URL Status code Length
01:12:54 24 Sept ... HTTP Request POST http://127.0.0.1:5000/
Request
Pretty Raw Hex
POST / HTTP/1.1
Host: 127.0.0.1:5000
Content-Length: 59
Cache-Control: max-age=0
sec-ch-ua: "Chromium";v="145", "Not:A-Brand";v="99"
sec-ch-ua-mobile: ?0
sec-ch-ua-platform: "Linux"
Accept-Language: en-GB,en;q=0.9
Origin: http://127.0.0.1:5000
Content-Type: application/x-www-form-urlencoded
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Sec-Fetch-Site: same-origin
Sec-Fetch-Mode: navigate
Sec-Fetch-User: ?1
Sec-Fetch-Dest: document
Referer: http://127.0.0.1:5000/
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
username=Shaikh&message=Hello%2C+This+is+my+Experiment+No+9
Search
0 highlights
Inspector
Request attributes 2
Request query parameters 0
Request body parameters 2
Request cookies 0
Request headers 19
Event log (1)
All issues
Memory: 130.9MB of 3.82GB
Disabled
SSD Lab - Burp Suite
127.0.0.1:5000
Burp Suite Traffic Demo
Username
Message
Send Request
{% if username %}
Server Response
Username: BurpTest
Message: Hello, This is my Experiment No 9 {% endif %}